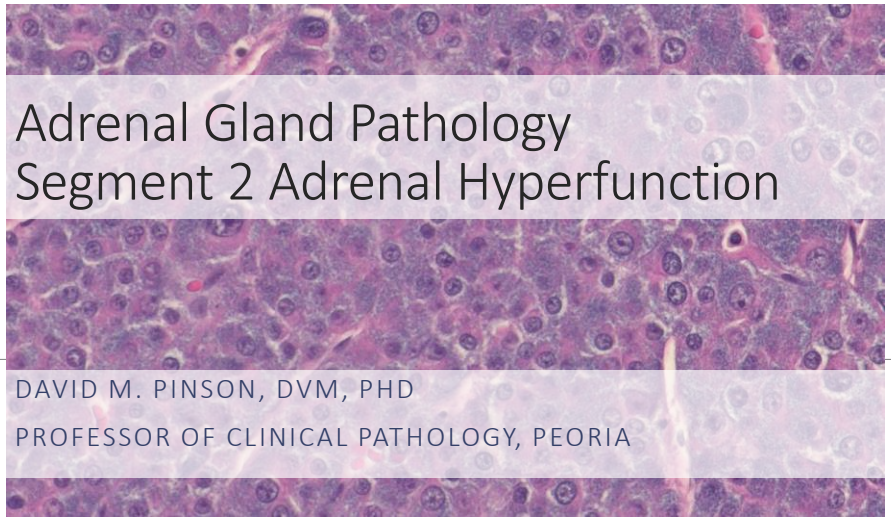


Adrenal Gland Pathology
Segment 2 Adrenal Hyperfunction

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Greetings students and welcome to the second recording in this series on adrenal gland disease.

Microlecture Series

Segment	Title	Duration
Segment 1	Overview	4 min 46 sec
Segment 2	**Adrenal Hyperfunction**	**17 min 13 sec**
Segment 3	Adrenal Hypofunction	14 min 35 sec
Segment 4	Adrenal Neoplasia	10 min 01 sec

This is the second recording in a series of four.

Learning Objectives

Goal: Identify the regulation and control mechanisms of adrenocortical function and describe the approaches for pharmacological intervention

Goal: Specify the biochemistry, physiology and pathology of the catecholamines and the stimuli that modulate adrenomedullary function

LO1 Evaluate the epidemiology, risk factors, and signs and symptoms of hypercortisolism.

LO2 Describe the etiology, clinical features, pathogenesis, pathophysiology, prognosis and treatment options of Cushing's disease.

LO3 Identify the causes and consequences of oversecretion of adrenal androgens; describe the basis for adrenal virilizing disorders.

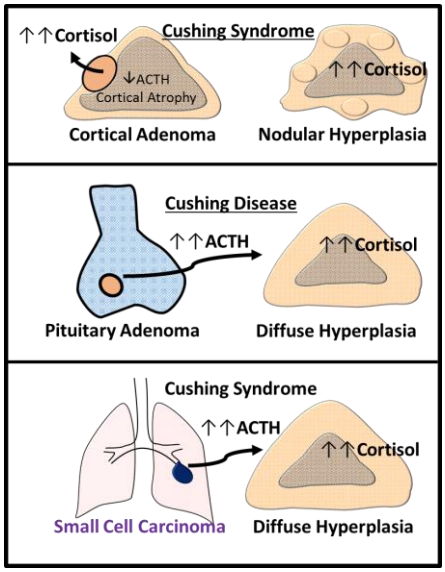
This recording addresses diseases of adrenal hyperfunction, essentially too much hormone. The learning objectives deal predominantly with cortisol, but aldosterone can also be affected. Sex steroids can be affected. I'll address these three learning objectives plus a few other diseases.

LO1
LO2

Cushing Syndrome

Defined: Excess glucocorticoid; any cause

- Most Common Cause: Therapeutic steroids
- Endogenous hypercortisolism
  - ACTH Independent
    - Adrenal cortical adenoma – 10%
    - Adrenocortical carcinoma – 5%
    - Primary adrenal cortical hyperplasia – 5%
  - ACTH Dependent
    - ACTH from Pituitary – 70% (**Cushing Disease**)
    - Paraneoplastic ACTH – 10%
      (half are small cell carcinoma)
    - CRH producing tumors (rare)



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There are two important terms on this slide. They are Cushing syndrome and Cushing disease. They are similar, but not the same. These are problems of excess glucocorticoids. Cushing syndrome is a constellation of clinical signs due to excess glucocorticoids of any cause. That may be therapeutically administered glucocorticoids or endogenously produced glucocorticoids. The most common cause of Cushing syndrome is therapeutic steroid administration.

However, excess production of endogenous cortisol can produce a very similar clinical syndrome. Excess cortisol production can be **independent** of ACTH or directly driven by excess ACTH from the pituitary or other sources. The ACTH **independent** syndromes may be due to functional tumors or idiopathic adrenal hyperplasia that is not regulated by ACTH. In such cases, the hypothalamus reads excess cortisol and shuts off ACTH so that ACTH values are very low. For a functional tumor that produces hormone, the surrounding gland will become atrophic because ACTH is low.

The ACTH **dependent** syndromes can be due to two dominant problems. The first is a pituitary tumor producing too much ACTH. This constitutes the formal Cushing Disease. The most common cause is a pituitary microadenoma. In this scenario, ACTH will be elevated and the adrenal cortex will be diffusely hyperplastic. The

second scenario is a neuroendocrine tumor somewhere producing ACTH. The most common tumor is small cell carcinoma of the lung, but other neuroendocrine tumors can do this. This a paraneoplastic syndrome. This unregulated production of ACTH drives diffuse adrenal hyperplasia. A rare syndrome of excess corticotropin releasing hormone is also recognized.

LO1
LO2

Cushing Syndrome Clinical Features

- Central truncal obesity
- Fat redistribution ("buffalo hump")
- Round (moon) facies
- Weakness, fatigue
- Hypertension (cortisol = mildly mineralocorticoid)
- Glucose intolerance
- Diabetes
- Menstrual disturbances (women)
- Loss of collagen
  - Bone resorption → Osteoporosis
  - Thinning of skin → Striae
- Occasionally mineralization of collagen (calcinosis cutis)

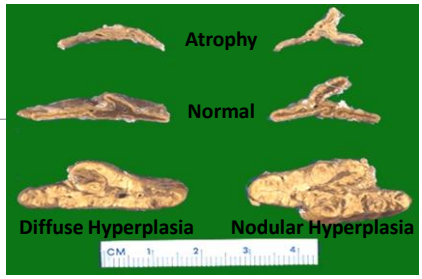


What does Cushing syndrome look like? There is a constellation of clinical features that may raise suspicion of this problem. Fat will be redistributed centrally and may result in an accumulation on the back, termed a buffalo hump. The face becomes rounded, which is termed "moon facies". Patients may also experience fatigue and glucose intolerance or frank diabetes. Since cortisol has some mineralocorticoid effects, hypertension may result. While this sounds a little like metabolic syndrome, it is different. There is a loss of collagen and this results in thin skin. Because of collagen loss, osteoporosis becomes a problem. Striae may develop on the abdomen. With long term steroid excess, the dermal collagen may mineralize in a condition called calcinosis cutis. The photographs illustrate some of these changes and the striking resolution after treatment.

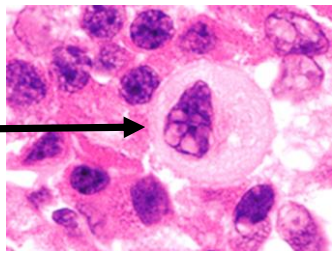
LO1
LO2

Cushing Syndrome Pathology

- Adrenal Gland
  - Exogenous steroid → Cortical atrophy
  - Endogenous excess steroid
    - Endogenous ACTH dependent→ diffuse hyperplasia
    - Endogenous ACTH independent → nodular hyperplasia
  - Functional cortical tumor: tumor nodule with surrounding cortical atrophy (not shown)
- Pituitary Gland
  - Microadenoma →ACTH production



ACTH Producer
Crooke hyaline change caused by high steroids; accumulation of keratin filaments



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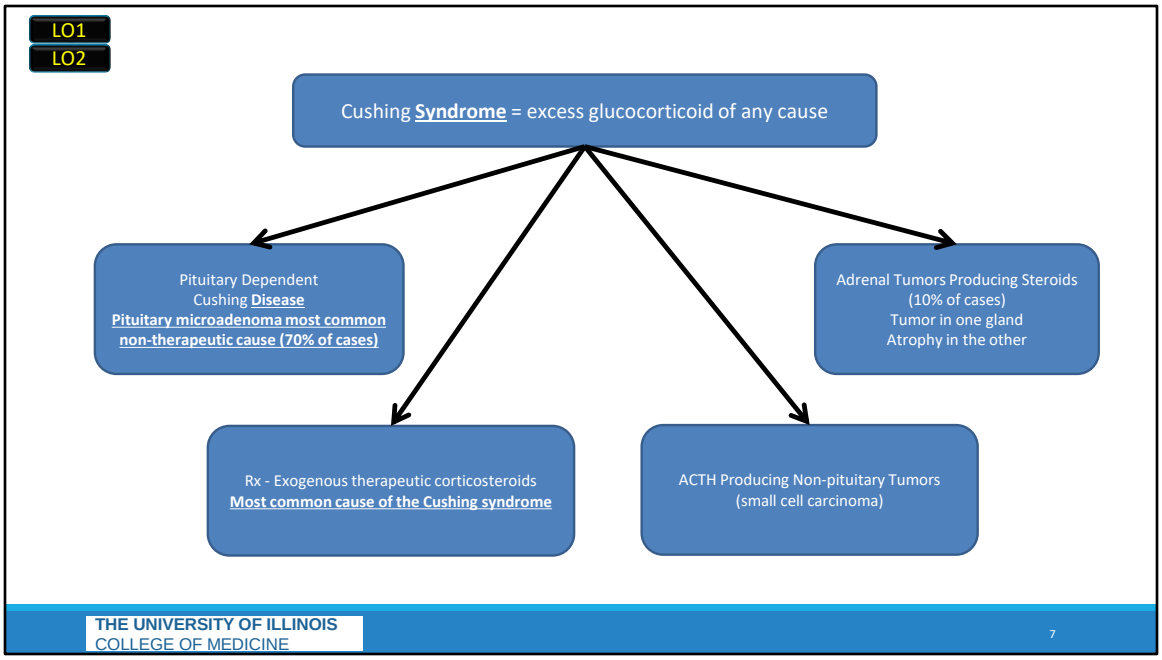
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The morphology of the adrenal gland depends on the pathogenesis of the steroid excess. If a patient is taking therapeutic glucocorticoids for any amount of time, the ACTH will be low because of that excess exogenous steroid. In this case, the adrenal cortex may become atrophic. This can be a significant problem if patients abruptly stop taking the drug. We'll learn about that problem in the third recording.

With a productive or functional adrenal cortical tumor, the surrounding cortex may be atrophic due to lack of ACTH.

In contrast, with a pituitary microadenoma producing ACTH or a neuroendocrine tumor producing ACTH, the cortex may be diffusely hyperplastic. The photomicrograph shows a high power image of a pituitary microadenoma with Crooke hyaline induced by excess steroids.

In the next few slides, I'll discuss the laboratory testing of these scenarios and what to expect from classical cases of hypercortisolism.



Cushing syndrome is excess glucocorticoid of any cause. There are four classical syndromes that can produce this problem.

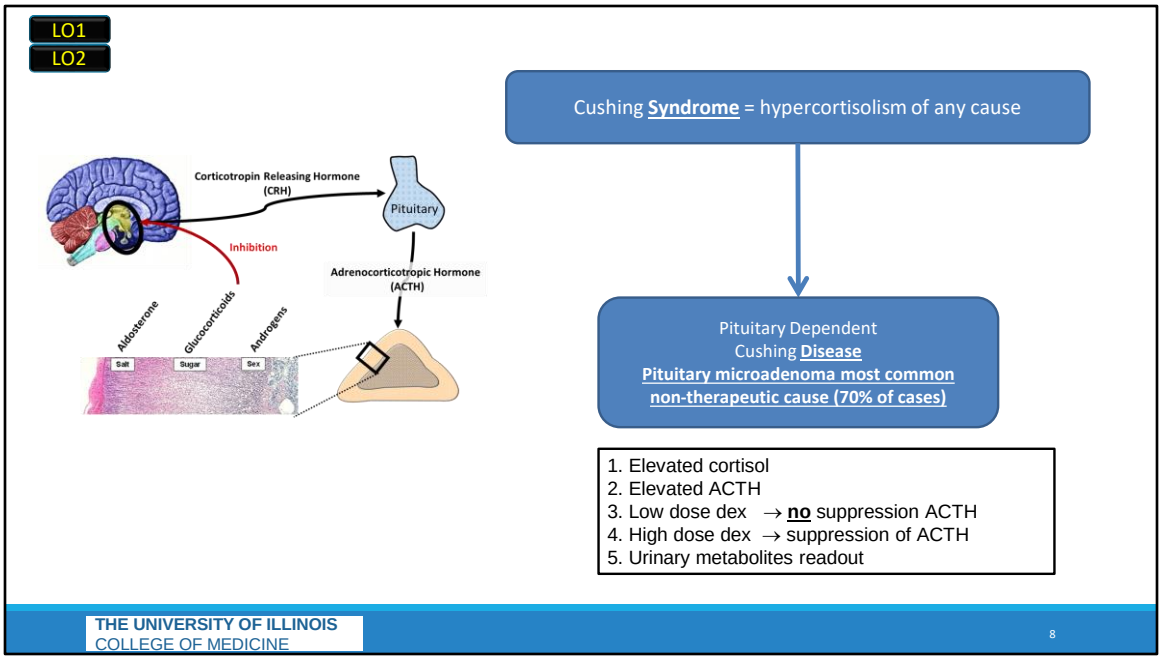
The first is Cushing Disease due to a pituitary micro-adenoma secreting ACTH.

The second is exogenous therapeutic steroids.

The third is a non-pituitary neuroendocrine tumor producing ACTH.

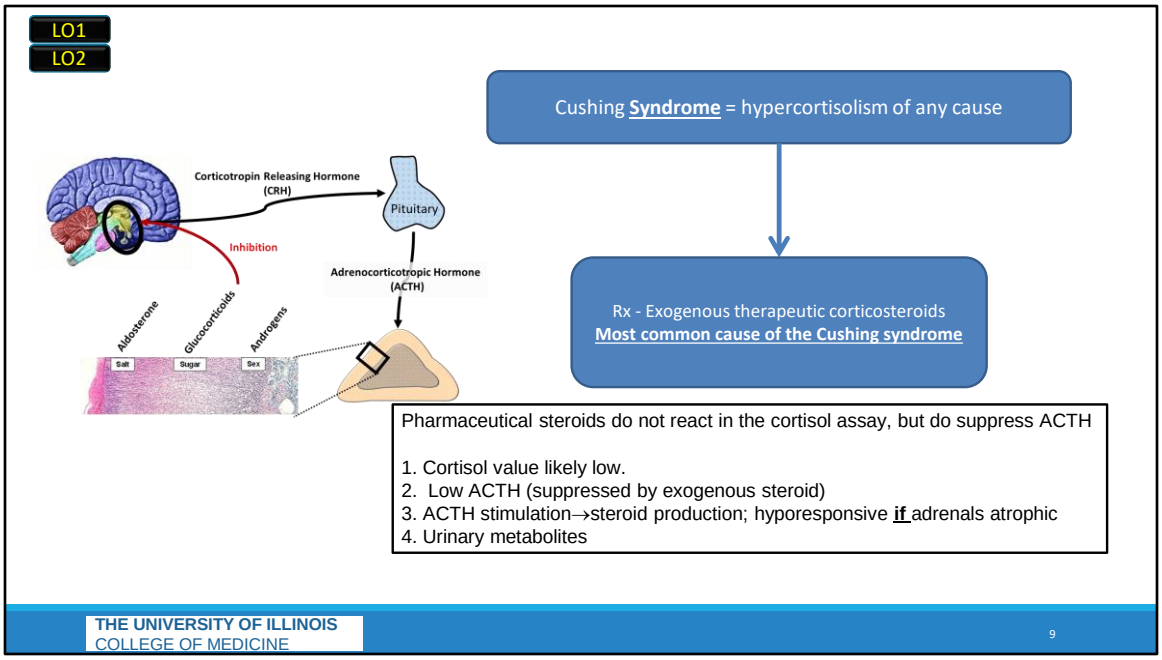
The last is an adrenal cortical tumor producing cortisol independent of ACTH.

Let's look at the laboratory findings for each of these.



For Cushing disease, there is a pituitary tumor producing too much ACTH. This drives the adrenal cortex to produce too much cortisol. The laboratory findings to some degree are intuitive. You expect elevated ACTH and elevated cortisol, perhaps with loss of diurnal rhythms. What time of day you measure cortisol matters for the reference range. For this problem, there are two variations on a test called the dexamethasone suppression test. In this test, dexamethasone is given in either low dose or high dose. In a normal person, dexamethasone will easily suppress ACTH production. The low dose dex suppression test does not usually suppress the production of ACTH by a pituitary microadenoma, but the high dose test will usually shut off ACTH production by those tumors. This is not always the case, but usually. The readout is steroid metabolites in the urine. One could measure blood cortisol as well, but urinary metabolites gives a more comprehensive idea of steroid production over a period of time rather than a moment in time for serum cortisol.

So the low and high dose dex suppression is used to distinguish pituitary dependent Cushing disease from other Cushing syndromes.

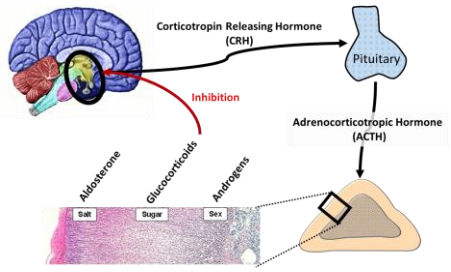


The second scenario is therapeutically administered steroids. The synthetic steroids usually don't react in the assay for cortisol, but these synthetic glucocorticoids will suppress ACTH production. Because ACTH production is suppressed, cortisol will be low since the adrenal gland is not stimulated to produce the hormone. If the therapeutic steroid is given long enough. The adrenal gland may become atrophic.

In this scenario, cortisol is expected to be low. ACTH is expected to be low because the exogenous steroids suppress ACTH production. Urinary metabolites would depend on which drug is given.

There is another test called the ACTH stimulation test. This is typically used when adrenal deficiency is suspected. It can be used here to determine if the adrenal gland is able to respond after long term exogenous steroids. If the adrenal gland is atrophic, it may not respond sufficiently to ACTH. In such cases, serum cortisol after ACTH would remain low indicating a nonresponsive cortex. This glandular atrophy is why these drugs are usually tapered off to allow the adrenal gland time to recover.

LO1
LO2



Cushing **Syndrome** = hypercortisolism of any cause

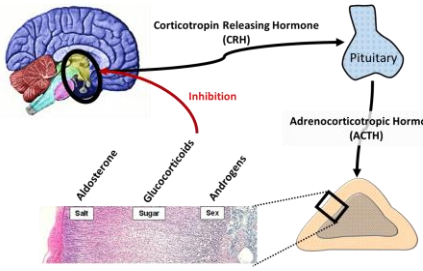
ACTH Producing Non-pituitary Tumors (small cell carcinoma)

1. Elevated cortisol
2. Elevated ACTH
3. Dex suppression → **no** suppression at any dose
4. Urinary metabolites

The third scenario is a neuroendocrine tumor producing ACTH and driving hyperplasia of the adrenal cortex. This results in excess cortisol and Cushing syndrome. Here, we expect ACTH to be high as a paraneoplastic syndrome. Cortisol will be high. This is another case where the dexamethasone suppression test is important to distinguish pituitary dependent disease from a paraneoplastic syndrome. ACTH would be high, but its production by a neuroendocrine tumor would not be suppressed at any dose of dexamethasone. Urinary cortisol metabolites is the readout.

LO1
LO2

Cushing **Syndrome** = hypercortisolism of any cause



Adrenal Tumors Producing Steroids
(10% of cases)
Tumor in one gland
Atrophy in the other

1. Elevated cortisol
2. Low ACTH (feedback suppression)
3. Dex suppression test not relevant
4. Urinary metabolites

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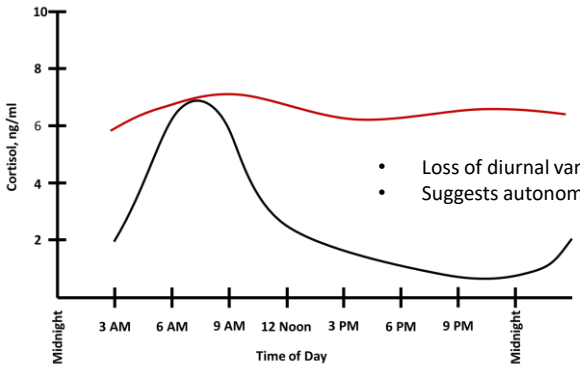
11

The final scenario is that of functional adrenal tumor producing excess steroid hormone. Here, the excess steroid suppresses the normal production of ACTH by the pituitary. We would expect ACTH to be low by this feedback suppression. Cortisol would be high. Dexamethosone suppression will not tell you anything since ACTH is already low. You cannot suppress it any further. It is a meaningless test in the context of low ACTH. The readout is urinary metabolites.

All of these laboratory tests are built upon standardized drug doses, standardized sampling times, and reference ranges, usually in detailed assay protocols. The tests usually specify when stimulatory or suppressive drugs are given and when readout samples are collected and how. Remember the diurnal rhythms.

LO1
LO2

Cortisol Concentrations are Diurnal



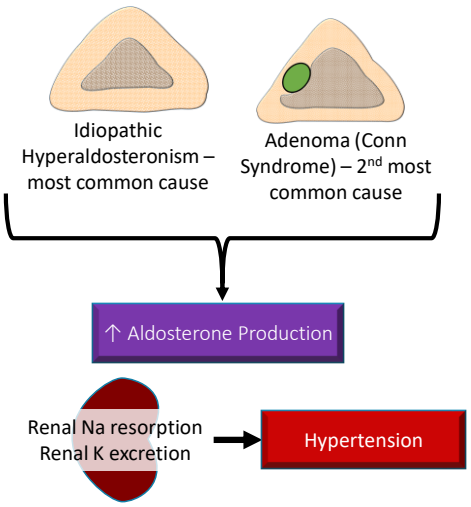
- Loss of diurnal variation is clinically significant
- Suggests autonomous secretion

In fact, a loss of diurnal rhythm is clinically significant. We expect cortisol to surge overnight and peak around 8 am in the morning. The loss of that rhythm or a consistently elevated serum cortisol can produce a Cushing phenotype. Thus, if you measure cortisol as high in the afternoon or evening but normal value at 8 AM, that may be very meaningful. However, urinary metabolites give a readout of just how much steroid is produced, metabolized and excreted over time. It is a broader spectrum of measurement. The loss of diurnal rhythm suggests autonomous production due to one of these causes.

Hyperaldosteronism

Primary Hyperaldosteronism

- Adrenal origin aldosterone
- Idiopathic zona glomerulosa hyperplasia – 60%
- Cortical adenoma (Conn Syndrome) – 35%
- Cortical carcinoma – rare
- Glucocorticoid remediable hyperaldosteronism
  - Rare disease
  - CYP11B2 gene for aldosterone synthesis under ACTH control
  - ACTH drives aldosterone secretion
  - Inhibition of ACTH →inhibition of aldosterone secretion

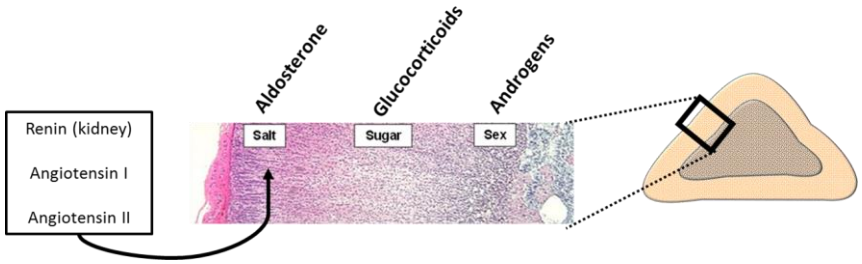


There is another variation on excessive production of hormone by the adrenal cortex. This is hyperaldosteronism. Hyperaldosteronism can be primary or secondary. This almost always translates to hypertension because of volume expansion. This can be due to excessive production by hyperplastic zona glomerulosa, probably in 60% of cases. It can be due to an adenoma of the zona glomerulosa, a disease termed Conn syndrome. Cortical carcinomas producing aldosterone are rare. There is an oddball chromosomal lesion that brings aldosterone production under the control of ACTH. This is called glucocorticoid remediable hyperaldosteronism. This means that glucocorticoids suppress ACTH and this mitigates both aldosterone concentration and hypertension.

Hyperaldosteronism

Secondary Hyperaldosteronism

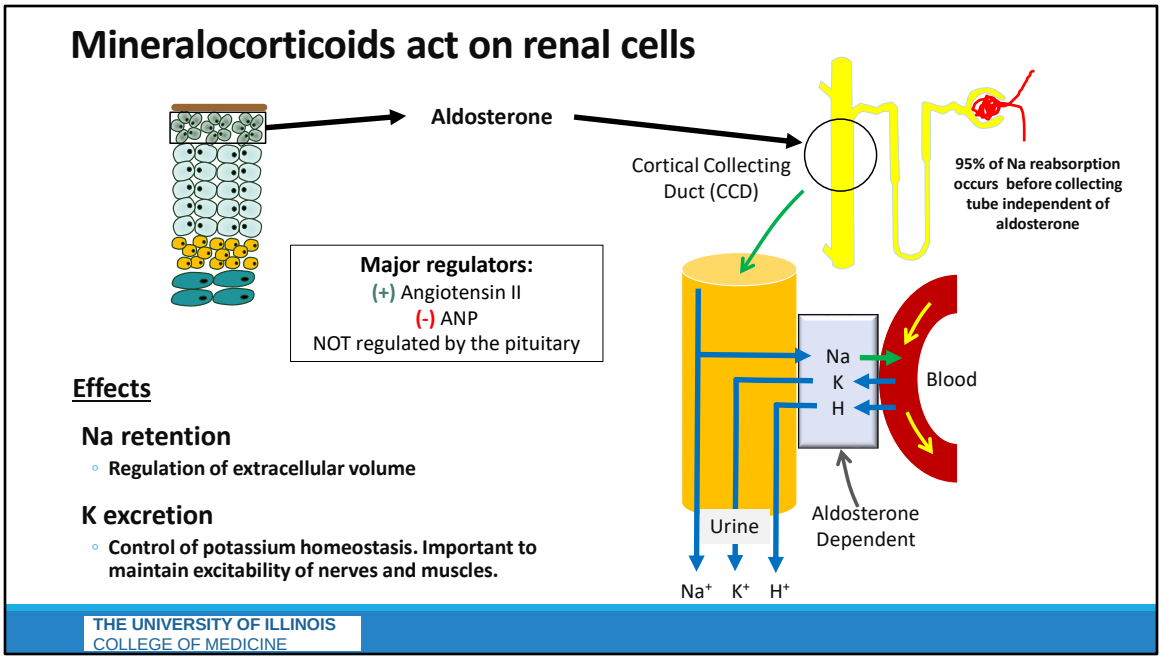
- Extra adrenal driver for aldosterone secretion
- Renin driven via reduced renal perfusion
- Renin driven via ineffective circulating volume
- Pregnancy



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The concentration of aldosterone in serum is a bit challenging to interpret. Why? Because there are secondary forms of hyperaldosteronism. Most of them are the result of decreased perfusion of the kidney and this causes renin secretion. Renin converts angiotensinogen to angiotensin 1. Angiotensin 1 is converted to angiotensin 2 by converting enzyme in the lung…the ACE enzyme. Angiotensin 2 is a vasoconstrictor and stimulates the adrenal to produce aldosterone. Why is this challenging? Because renin and aldosterone have to be interpreted in the context of volume status and this can be a very complicated picture. Expanded volume typically results in hypertension unless the heart is failing. Narrowing of the renal arteries can result in under-perfusion of the kidney and renin production in the face of normal volume. So beware of the complexities of aldosterone and renin concentration values. Remember volume status. It is important.



You need to remember your physiology and what aldosterone does. It manages and regulates sodium, potassium and hydrogen ions, mostly in the collecting ducts of the kidney. The vast majority of sodium re-absorption in the kidney is outside of aldosterone control. Only about 3% of sodium reabsorption in the collecting duct is regulated by aldosterone. This is one of your physiology slides to remind you of these parameters.

Aldosterone's classical functions are twofold: regulation of extracellular volume and control of potassium homeostasis.

95% of Sodium in reabsorbed before the distal nephron independently of aldosterone.
The reabsorption of the 5% of sodium remaining in urine can be regulated to control salt and volume homeostasis.

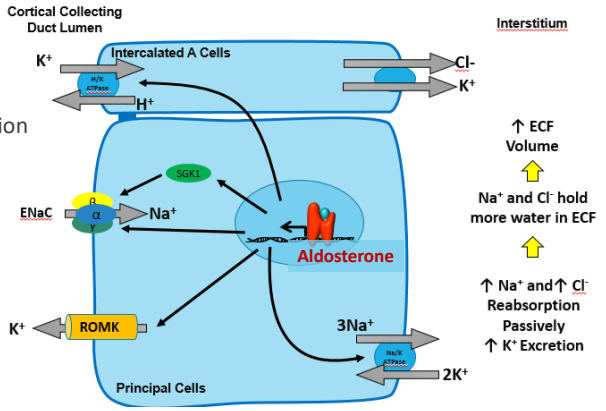
Potassium excretion regulated by aldosterone is important for maintaining excitability of nerves and muscles.

As a reminder, aldosterone synthesis and secretion is not regulated by the pituitary.

Hyperaldosteronism

Clinical Expression and Diagnosis

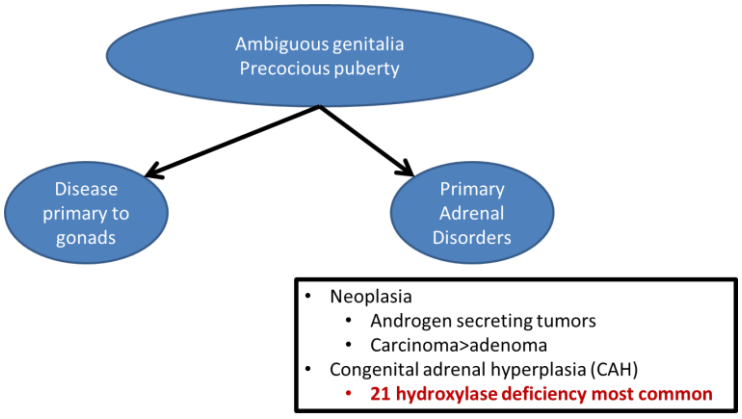
- **90% of cases expressed as hypertension**
- Na^+ retention; K^+ excretion → volume expansion
- Hypokalemia (classically, but not always)
- Laboratory testing
  - Serum electrolytes
  - Aldosterone/renin ratio
  - Aldosterone suppression test (Na^+ loading)
  - Urine electrolytes
  - Urine anion gap



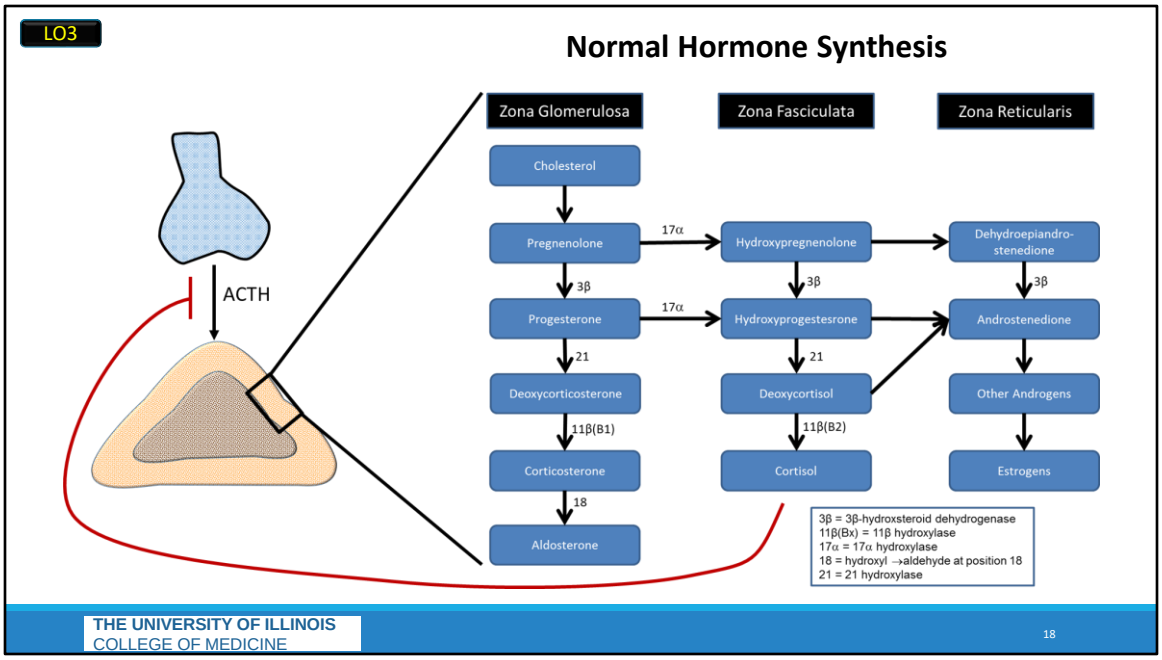
Aldosterone controls sodium reabsorption, K excretion, and hydrogen ion excretion. If these three ions are askew, consider that aldosterone may be involved. The laboratory testing demands that you have some understanding of the patient's volume status to interpret those values correctly. In the context of expanded volume, aldosterone should normally be low. Sodium loading should reduce aldosterone secretion since you want to excrete excess sodium. Remember that antidiuretic hormone may play a role in the sodium concentration.

LO3

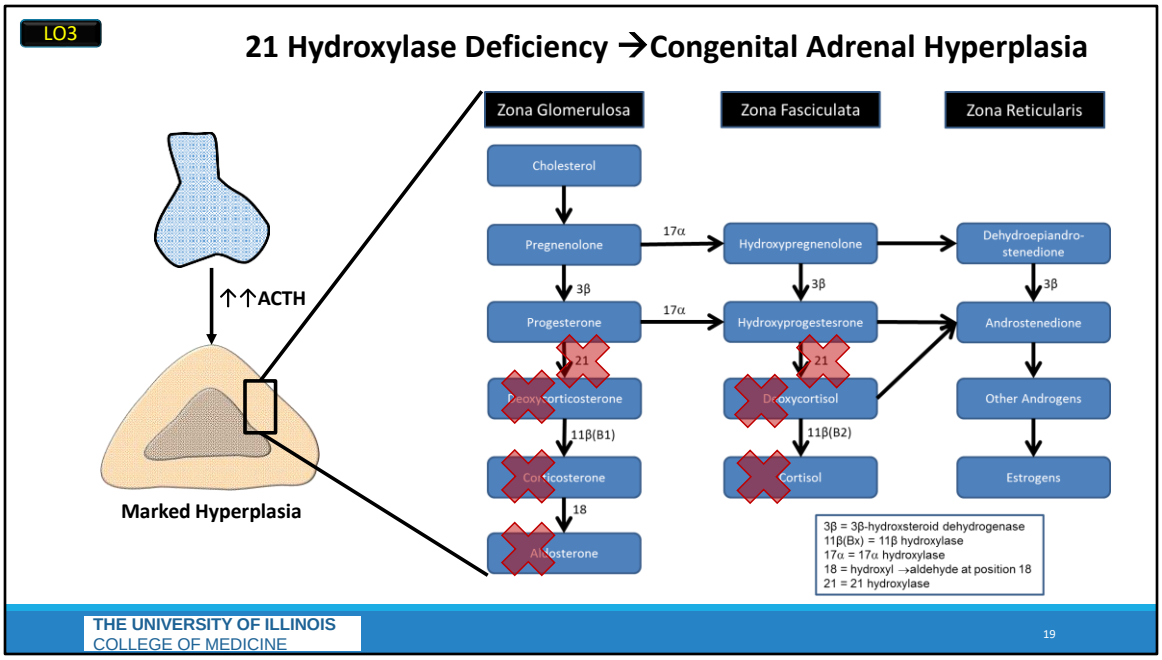
Disorders of Sexual Differentiation



The last topic of this recording is the subject of congenital adrenal hyperplasia, which is a virulizing syndrome that can result in ambiguous genitalia. This problem of ambiguous genitalia can be due to adrenal disease or primary gonadal disease. We are talking here about the adrenal gland.



This is where these synthetic pathways for hormones becomes important. So each zone of the adrenal gland has the enzymes required to manufacture specific hormones. Cortisol produced by the zona fasciculata feeds back to the hypothalamus and pituitary to regulate ACTH. I commented in the first recording that you should note 21 hydroxylase.



If you take out 21 hydroxylase by mutation, this essentially takes out production of aldosterone and cortisol. Since the androgen machinery remains intact, metabolites are shunted to the androgen pathway. In the 21 hydroxylase mutation, androgens are produced in excess. To complicate the problem, cortisol production is lost. This causes a problem because there is no feedback inhibition of the hypothalamus and pituitary. The hypothalamus reads this a deficient cortisol and stimulates the pituitary to produce ACTH in high amounts. This drives adrenal hyperplasia, which can be massive. However, due to the enzyme deficiency, the adrenal gland cannot manufacture cortisol. So there is a constant ACTH stimulation of the adrenal gland leading to hyperplasia. The 21 hydroxylase deficiency is the most common problem and it is a prominent USMLE board topic.

LO3

Congenital Adrenal Hyperplasia

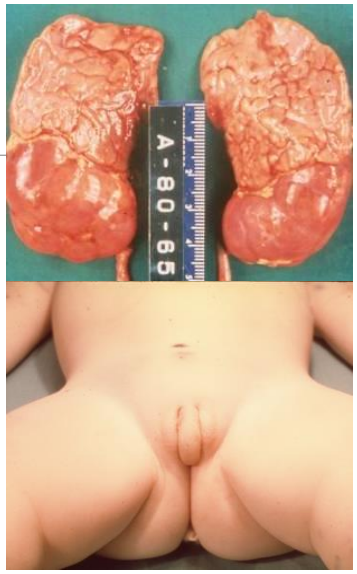
- Cause
  - Autosomal recessive
  - >90% of case due to missing 21-hydroxylase
  - Precursors shunted to androgens
    - ↑ androgens
    - ↓ glucocorticoids
    - ↓ mineralocorticoids
- Manifestations – 3 syndromes
  - Salt wasting type
  - Virulization with some mineralocorticoids
  - Non-classic
    - Partial deficiency
    - Hirsutism
    - Menstrual irregularity
- Rx
  - Glucocorticoids
  - Mineralocorticoids
  - Suppress ACTH

The deficiency is inherited as autosomal recessive. 21 hydroxylase deficiency constitutes 90% of the cases. There are three categories of clinical expression. They are salt wasting due to no aldosterone, virulization due to androgen production, and a nonclassical type with partial deficiency. This latter category is usually recognized at a later age rather than infancy. Therapy is directed at restoration of glucocorticoids that suppress ACTH and mineralocorticoids that restore sodium, potassium and hydrogen ion regulation.

LO3

Congenital Adrenal Hyperplasia

- Morphology
  - Pituitary corticotroph hyperplasia
  - Adrenal cortical hyperplasia
  - Virulization
    - Girls – virulization, ambiguous genitalia
    - Boys – precocious puberty
  - Salt Wasting Syndromes
    - Hyponatremia
    - Hyperkalemia
    - Acidosis
    - Hypotension
    - Cardiovascular collapse, death



The typical lesions of congenital adrenal hyperplasia include pituitary hyperplasia, massive adrenal hyperplasia, and ambiguous genitalia. The salt wasting syndromes are typical of missing aldosterone. We will talk more about the absence of aldosterone in the next recording.

Summary

- The causes of adrenal corticosteroid excess.
- The distinction of Cushing syndrome and Cushing disease.
- The clinical presentation of hypercortisolism.
- The laboratory testing for hypercortisolism.
- The biology of hyperaldosteronism and the most common causes.
- The pathogenesis and clinical expression of virulizing syndrome secondary to 21 hydroxylase deficiency.
- Up Next: Adrenal Hypofunction

In this session, I've covered the syndromes of cortisol and glucocorticoid excess. I have distinguished Cushing syndrome from Cushing disease. I summarized the clinical features of cortisol and glucocorticoid excess, aldosterone excess, and virulizing syndromes. In the next recording, we'll cover adrenal gland hypofunction.